



Course: Linear Algebra and Optimization

Course Code: 23MA4BSLAO

Unit – 1: Continuous Optimisation-1

Gradient of scalar with respect to vector

Let $\phi = \phi(x_1, x_2, \dots, x_n)$ be the scalar function and $x = [x_1 \ x_2 \ \dots \ x_n]$ be the row vector then gradient of ϕ w.r.t. x is $\frac{d\phi}{dx} = \left[\frac{d\phi}{dx_1} \ \frac{d\phi}{dx_2} \ \dots \ \frac{d\phi}{dx_n} \right]$. (Numerator Layout)

Let $\phi = \phi(x_1, x_2, \dots, x_n)$ be the scalar function and $X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ be the column vector then

gradient of ϕ w.r.t. X is $\frac{d\phi}{dX} = \begin{bmatrix} \frac{d\phi}{dx_1} \\ \frac{d\phi}{dx_2} \\ \vdots \\ \frac{d\phi}{dx_n} \end{bmatrix}$. (Denominator Layout)

Functions of several variables and partial differentiation

Examples:

Find the gradient of following functions at the given point:

1. $u = e^{xyz}$ and $p = (1, 1, 1)$

2. $f = xyz$ and $p = (1, 0, 1)$

3. $g = x^2 y^2 z$ and $p = (2, 3, -1)$

4. $g = 3x^2 + y^2 + z^2$ and $p = (0, 0, 2)$

5. $g = 3xe^{2y} + e^z y^2 + z^2$ and $p = (1, 0, 0)$

6. $f = 2x^3 e^{y^2} z^2$ and $p = (2, 0, 3)$

Jacobian Matrix

The **Jacobian matrix** is a matrix composed of the first-order partial derivatives of a multivariable function. Let $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be defined by $f(x_1, x_2, \dots, x_n) = (f_1, f_2, \dots, f_m)$ then Jacobian of f is defined as

$$J_X(f) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \dots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \dots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \dots & \frac{\partial f_m}{\partial x_n} \end{bmatrix} = \begin{bmatrix} \frac{df_1}{dx} \\ \frac{df_2}{dx} \\ \vdots \\ \frac{df_n}{dx} \end{bmatrix}$$

Examples:

Find the Jacobian matrix of the following function:

1. $f(x, y) = (x^4 + 3y^2x, 5y^2 - 2xy + 1)$ at $(1, 2)$.

2. $f(x, y) = (e^{xy} + y, y^2x)$ at $(0, -2)$.

3. $f(x, y) = (x^3y^2 - 5x^2y^2, y^6 - 3y^3x + 7)$ at $(1, -1)$.



4. $f(x, y, z) = \left(z \tan(x^2 - y^2), xy \ln\left(\frac{z}{2}\right) \right)$ at $(2, -2, 2)$.
5. $f(x, y, z) = \left(xe^{2y} \cos(-z), (y - 2)^3 \sin\left(\frac{z}{2}\right), e^{2y} \ln\left(\frac{x}{3}\right) \right)$ at $(3, 0, \pi)$.

Local and Global Optima

Definition: Let $X \subset \mathbb{R}^2$, let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function, and let $p \in X$.

1. f has a local maximum at P if there exist neighbourhood U of P in X such that $f(P) \geq f(X)$ for all $X \in U$.
2. f has a local minimum at P if there exist neighbourhood U of P in X such that $f(P) \leq f(X)$ for all $X \in U$.
3. f has a global maximum at P if $f(P) \geq f(X)$ for all $X \in U$, in which case we call $f(p)$ the global maximum value of f .
4. f has a global minimum at P if $f(P) \leq f(X)$ for all $X \in U$, in which case we call $f(p)$ the global minimum value of f .

Second derivative test: Let $z = f(x, y)$ be the differentiable function. For a critical point (a, b) , let

$$D = f_{xx}(a, b)f_{yy}(a, b) - (f_{xy}(a, b))^2$$

be the discriminant.

- If $D > 0$ and $f_{xx}(a, b) > 0$, then f has a local minimum at (a, b) .
- If $D > 0$ and $f_{xx}(a, b) < 0$, then f has a local maximum at (a, b) .
- If $D < 0$, then (a, b) is the saddle point of f .
- If $D = 0$, then the test gives no information.

To find optimum point of multivariate functions more than two variables, we need to compute its Hessian matrix at all the critical points. **Hessian matrix** of function of three variable $f(x, y, z)$ is given by

$$H = \begin{bmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{bmatrix}$$

Second derivative test: Find the critical points, plug them in the Hessian matrix, and compute their eigenvalues.

- If all the eigenvalues are strictly positive, then the critical point is a local minimum.
- If all the eigenvalues are strictly negative, then the critical point is a local maximum.
- If H has both positive and negative eigenvalues and all are non-zero, then the critical point is saddle point.
- If any of the eigenvalue is zero, then the test gives no information.

Examples:

For the following multivariate function:

- i) Find all the stationary points of the function.
- ii) Find the Hessian matrix.
- iii) Classify the stationary points.

- | | |
|--|---|
| 1. $f(x, y) = x^2 + y^2 - 4x + 2y$ | 5. $f(x, y, z) = x^2 + y^2 + z^2$ |
| 2. $f(x, y, z) = xy + yz + zx - 4x + 2y$ | 6. $f(x, y, z) = 2x^2 + 3y^2 + 4z^2 - 4xyz$ |
| 3. $f(x, y, z) = x^2 + y^2 + z^2 + xy + yz + zx$ | 7. $f(x, y, z) = 3x^2 + 2y^2 + 5z^2 - 6xyz$ |
| 4. $f(x, y, z) = -x^3 + 3xz + 2y - y^2 - 3z^2$ | 8. $f(x, y, z) = x^3 + y^2 + z^2 - 3xyz$ |



Gradient of vector with respect to vector

If $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ then the gradient of f is

$$\frac{df}{d\mathbf{X}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} = \begin{bmatrix} \frac{df_1}{d\mathbf{X}} \\ \frac{df_2}{d\mathbf{X}} \\ \vdots \\ \frac{df_n}{d\mathbf{X}} \end{bmatrix} = J_{\mathbf{X}}(f)$$

Gradient of matrix with respect to vector

Let $f = \begin{bmatrix} f_{11} & f_{12} & \cdots & f_{1n} \\ f_{21} & f_{22} & \cdots & f_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ f_{m1} & f_{m2} & \cdots & f_{mn} \end{bmatrix}$ and $\mathbf{X} = [x_1 \ x_2 \ \cdots \ x_n]$ then gradient of f w.r.t. $\mathbf{X}$ is

$$\frac{df}{d\mathbf{X}} = \begin{bmatrix} \frac{df_{11}}{d\mathbf{X}} & \frac{df_{12}}{d\mathbf{X}} & \cdots & \frac{df_{1n}}{d\mathbf{X}} \\ \frac{df_{21}}{d\mathbf{X}} & \frac{df_{22}}{d\mathbf{X}} & \cdots & \frac{df_{2n}}{d\mathbf{X}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{df_{m1}}{d\mathbf{X}} & \frac{df_{m2}}{d\mathbf{X}} & \cdots & \frac{df_{mn}}{d\mathbf{X}} \end{bmatrix}$$

Note: Each element in the above matrix is of size $1 \times n$.

Gradient of matrix with respect to matrix

Let $f = \begin{bmatrix} f_{11} & f_{12} & \cdots & f_{1n} \\ f_{21} & f_{22} & \cdots & f_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ f_{m1} & f_{m2} & \cdots & f_{mn} \end{bmatrix}$ and $\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{bmatrix}$ then gradient of f w.r.t. $\mathbf{X}$ is

$$\frac{df}{d\mathbf{X}} = \begin{bmatrix} \frac{df_{11}}{d\mathbf{X}} & \frac{df_{12}}{d\mathbf{X}} & \cdots & \frac{df_{1n}}{d\mathbf{X}} \\ \frac{df_{21}}{d\mathbf{X}} & \frac{df_{22}}{d\mathbf{X}} & \cdots & \frac{df_{2n}}{d\mathbf{X}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{df_{m1}}{d\mathbf{X}} & \frac{df_{m2}}{d\mathbf{X}} & \cdots & \frac{df_{mn}}{d\mathbf{X}} \end{bmatrix}$$

Note: Each element in the above matrix is of size $m \times n$.

Useful identities for computing Gradient

- | | |
|--|--|
| 1. $\frac{\partial(\text{Trace}(f(\mathbf{X})))}{\partial \mathbf{X}} = \text{Trace}\left(\frac{\partial f(\mathbf{X})}{\partial \mathbf{X}}\right)$ | 4. $\frac{\partial a^T \mathbf{X}^{-1} b}{\partial \mathbf{X}} = -(\mathbf{X}^{-1})^T a b^T (\mathbf{X}^{-1})^T$ |
| 2. $\frac{\partial (f(\mathbf{X}))^T}{\partial \mathbf{X}} = \left(\frac{\partial f(\mathbf{X})}{\partial \mathbf{X}}\right)^T$ | 5. $\frac{\partial x^T a}{\partial x} = a^T = \frac{\partial a^T x}{\partial x}$ |
| 3. $\frac{\partial(\det(f(\mathbf{X})))}{\partial \mathbf{X}} = \det(f(\mathbf{X})) \text{Trace}\left((f(\mathbf{X}))^{-1} \left(\frac{\partial f(\mathbf{X})}{\partial \mathbf{X}}\right)\right)$ | 6. $\frac{\partial a^T \mathbf{X} b}{\partial \mathbf{X}} = a b^T$ |

**Examples:**

1. Discuss the gradient of vector with respect to matrix.
2. Obtain the Gradient of vector $\mathbf{f} = [e^{x_0 x_1} \quad e^{x_2 x_3}]$ with respect to the matrix $\mathbf{x} = \begin{bmatrix} x_0 & x_1 \\ x_2 & x_3 \end{bmatrix}$.
3. Obtain the gradient of scalar $\phi = 4x_0 + 2x_1 - 3x_2 + x_4$ with respect to the matrix $\vec{x} = \begin{bmatrix} x_0 & x_1 \\ x_2 & x_3 \end{bmatrix}$.
4. Find the derivative of the vector $\begin{bmatrix} x_0 x_1 \\ x_0 + x_1 \\ x_2 \end{bmatrix}$ with respect to vector $[x_0 \quad x_1 \quad x_2]$ and hence find the nature of the resulting matrix at $[0,1,1]$.
5. Obtain the gradient of the vector $\begin{bmatrix} 4x_0 x_1 \\ 4x_0 + 2x_1 \\ 3x_2 \end{bmatrix}$ with respect to vector $[x_0 \quad x_1 \quad x_2]$ and hence find the values of x_0 and x_1 for which 8, 3 and 2 are the eigenvalues of derivative matrix.
6. Obtain the gradient of the matrix $\begin{bmatrix} x_0 x_1 \\ x_0 + e^{x_1} \\ x_2^2 \end{bmatrix}$ with respect to $\begin{bmatrix} x_0 \\ x_1 \\ x_2 \end{bmatrix}$.
7. Obtain the gradient of matrix $\mathbf{f} = \begin{bmatrix} \sin(x_0 + 2x_1) & 2x_1 + x_3 \\ 2x_0 + x_2 & \cos(2x_2 + x_3) \end{bmatrix}$ with respect to the matrix $\mathbf{x} = \begin{bmatrix} x_0 & x_1 \\ x_2 & x_3 \end{bmatrix}$.
8. Find the derivative of the matrix $\mathbf{f} = \begin{bmatrix} x_0^2 x_1 x_2 & x_1^2 x_2 x_3 \\ x_3^2 x_1 x_3 & x_2^3 x_1 \end{bmatrix}$ with respect to the matrix $\mathbf{x} = \begin{bmatrix} x_0 & x_1 \\ x_2 & x_3 \end{bmatrix}$.
9. Obtain the gradient of matrix $\begin{bmatrix} xyz & xy \\ zy & zx \\ z & 2yx \end{bmatrix}$ with respect to $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$.
10. Obtain the gradient of the matrix $\begin{bmatrix} xy & yz & zw \\ wx & wy & wz \end{bmatrix}$ with respect to $\begin{bmatrix} x & y \\ z & w \end{bmatrix}$.

Verification problems on gradient identities

1. Given $\mathbf{f} = \begin{bmatrix} x_0^2 x_1 x_2 & x_1^2 x_2 x_3 \\ x_3^2 x_1 x_3 & x_2^3 x_1 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} x_0 & x_1 \\ x_2 & x_3 \end{bmatrix}$. Verify
 - i. $\frac{\partial(\text{Trace}(\mathbf{f}(\mathbf{x})))}{\partial \mathbf{x}} = \text{Trace}\left(\frac{\partial \mathbf{f}(\mathbf{x})}{\partial \mathbf{x}}\right)$
 - ii. $\frac{\partial(\mathbf{f}(\mathbf{x})^T)}{\partial \mathbf{x}} = \left(\frac{\partial \mathbf{f}(\mathbf{x})}{\partial \mathbf{x}}\right)^T$
2. Given $\mathbf{f} = \begin{bmatrix} \sin(x_0 + 2x_1) & 2x_1 + x_3 \\ 2x_0 + x_2 & \cos(2x_2 + x_3) \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} x_0 & x_1 \\ x_2 & x_3 \end{bmatrix}$. Verify
 - i. $\frac{\partial(\text{Trace}(\mathbf{f}(\mathbf{x})))}{\partial \mathbf{x}} = \text{Trace}\left(\frac{\partial \mathbf{f}(\mathbf{x})}{\partial \mathbf{x}}\right)$
 - ii. $\frac{\partial(\mathbf{f}(\mathbf{x})^T)}{\partial \mathbf{x}} = \left(\frac{\partial \mathbf{f}(\mathbf{x})}{\partial \mathbf{x}}\right)^T$
3. Given $X = \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix}$, $a = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ and $b = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$. Verify $\frac{\partial(a^T X b)}{\partial \mathbf{x}} = ab^T$.
4. Given $X = \begin{bmatrix} x_1 & x_2 \\ x_3 & x_4 \end{bmatrix}$, $a = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ and $b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Verify $\frac{\partial(a^T X b)}{\partial \mathbf{x}} = ab^T$.
5. Given $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ and $a = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$. Verify $\frac{\partial(x^T a)}{\partial \mathbf{x}} = a^T$.



6. Given $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$ and $a = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 1 \end{bmatrix}$. Verify $\frac{\partial(a^T x)}{\partial x} = a^T$.

Convex sets and functions of separating hyperplanes

Convex set definition: A set C is convex if the line segment between any two points in C lies in C . That is, a set $C \subseteq \mathbb{R}^n$ is convex, if for all $x, y \in C$ and $\forall \lambda \in [0, 1]$, $\lambda x + (1 - \lambda)y \in C$.

A point of the form $\lambda x + (1 - \lambda)y$, $\lambda \in [0, 1]$ is called a convex combination of x and y . As λ varies between $[0, 1]$, a line segment is being formed between x and y .

Examples:

1. Show that intersection of two convex sets is a convex set. Is union of two convex sets is necessarily convex set. Justify your answer.
2. Show that the unit ball $B = \{u \in \mathbb{R}^n: ||u|| < 1\}$ is convex set.
3. Show that $S = \{(x_1, x_2): 2x_1 + 3x_2 = 7\} \in \mathbb{R}^2$ is a convex set.
4. Show that $S = \{(x_1, x_2): x_1^2 + x_2^2 \leq 4\}$ is a convex set.
5. Show that $S = \{(x_1, x_2, x_3): x_1^2 + x_2^2 + x_3^2 \leq 1\}$ is a convex set.
6. Show that $S = \{(x_1, x_2, x_3): 2x_1 - x_2 + x_3 \leq 4\}$ is a convex set.

Hyperplane

Hyperplane definition: For given scalar d in $\mathbb{R}$, the set of all $(x_1, x_2, \dots, x_n) \in \mathbb{R}^n$ such that $a_1x_1 + a_2x_2 + \dots + a_nx_n = d$ ($f(X) = d$) is called hyperplane. That is, if $\mathbf{n} = (a_1, a_2, \dots, a_n)$ and d is scalar then hyperplane is the set $H = \{X \in \mathbb{R}^n: \mathbf{n} \cdot X = d\}$.

Note: The vector $\mathbf{n}$ is normal to the hyperplane. Analyse How?

Definition: The hyperplane $H = [f: d]$ separates two sets A and B if one of the following holds:

- i. $f(A) \leq d$ and $f(B) \geq d$, or
- ii. $f(A) \geq d$ and $f(B) \leq d$.

If in the conditions above, all the weak inequalities are replaced by strict inequalities, then H is said to strictly separate A and B .

Examples:

1. Let $A = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ and $v = \begin{bmatrix} 1 \\ -6 \end{bmatrix}$, and let $H = \{X: A \cdot X = 12\}$. Find the implicit description of the parallel hyperplane $H_1 = H + V$.
2. Let $v_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$. Find an implicit description of hyperplane H that passes through v_1 and v_2 .
3. Let $v_1 = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$, $v_2 = \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix}$ and $v_3 = \begin{bmatrix} -1 \\ -2 \\ 5 \end{bmatrix}$. Find an implicit description of hyperplane H that passes through v_1, v_2 and v_3 .
4. Let $A = \left\{ \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} -1 \\ 6 \\ 0 \end{bmatrix} \right\}$, $B = \left\{ \begin{bmatrix} 0 \\ 5 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -3 \\ -2 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} \right\}$, and $\mathbf{n} = \begin{bmatrix} 3 \\ 1 \\ -2 \end{bmatrix}$. Find a hyperplane H with normal $\mathbf{n}$ that separates A and B . Is there any hyperplane that separates A and B strictly?